D H L A B

Semester Projects, Autumn 2026

Project sheets and how the projects articulate

Digital Humanities Laboratory, EPFL

17 sheets across five projects, with one articulation page per group

Compiled 11 September 2026 at 00:23, Europe/Zurich

Contents

Vintage Language Models

- **Rebuilding a Vintage LLM from scratch**
  Marc Younes
- **Verifying the cutoff and measuring what the model can do**
  Joe El Hayek
- **The futures the past imagined**
  Elio Khoury
- **The common sense the past never wrote down**
  Jitong Zhang
- **Conversations of the past**
  Iskandar Ghobril
- **The first rung of the rediscovery ladder**
  Diego Parra Ruiz
- **Building the Pastcasting Benchmark**
  Anaël Donini

Geolocating Artworks

- **Geolocating artworks from open museum collections**
  Benoît Meylan
- **Matching an artwork to the place it shows**
  Duc Anh Do
- **Placing the 1910 Paris flood back on the map**
  Camille Dupré-Tabti

Preparing Historical Data for Apertus

- **Measuring and cleaning the OCR**
  Yassine Dafer
- **The contamination and data-quality gate for the prepared corpus**
  Nastaran Hashemisanjani
- **The pivot between Apertus ingestion, Impresso, and the Time Atlas**
  Kaile Chu

EPFL Digital Twin Foundation Model

- **A foundation model for the campus**
  Ali Tahri Hassani
- **A multimodal model of the campus**
  Nacer Saadani-Hassani

IIIF Document Information Extraction

- **One door to a scattered heritage**
  Boheng Shang

- **Reading heritage with OCR**
  Hafidz Arsha

PROJECT GROUP

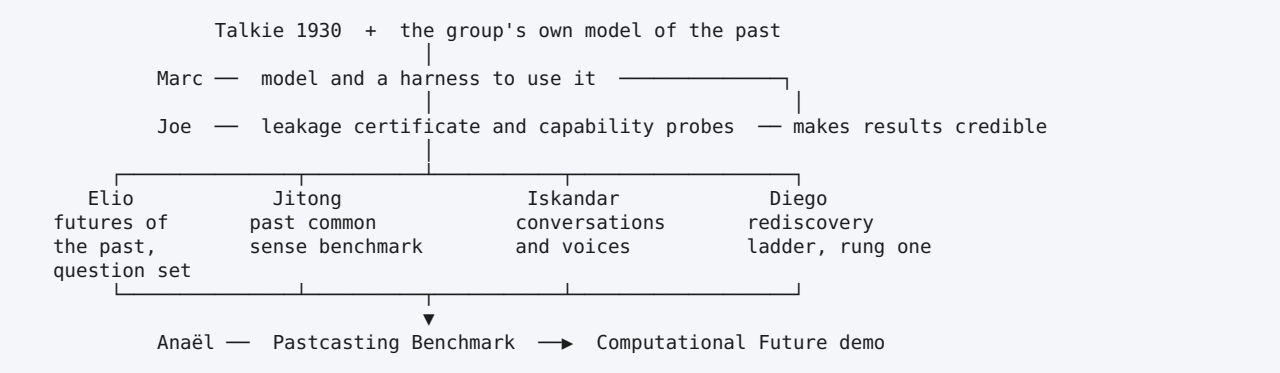
Vintage Language Models

7 sheets, preceded by the articulation page

How the Vintage Language Models projects articulate

DHLAB, autumn 2026. Supervision Frédéric Kaplan, Anaël Donini co-animates the Tuesday slot. One schematic page on how the Vintage projects fit together. Weekly slot, Tuesday 10:00 to 11:00, room INN 128.

The picture



The links

Marc builds the model and the harness, the tooling that makes a weak generator usable, so every other project queries the model through it. Joe audits the cutoff and certifies that the model did not see the future, and he measures what it can actually do, which is what makes any downstream result trustworthy. On that shared base each student opens one question of the field. Elio studies the futures the past imagined and produces the set of dated questions. Jitong builds the benchmark of past common sense. Iskandar gathers the attested dialogue and studies whose voice the model carries. Diego builds the rediscovery instrument, the case protocol and the hint ladder that measure whether a model of 1930 can reach a later discovery once it is handed the enabling analogy.

Anaël is the point of convergence. He takes the model and harness, the leakage certificate, the question set and two short vignettes, and assembles them into the Pastcasting Benchmark, whose first public instantiation is the demonstration at Computational Future. **The benchmark does not depend on any particular model**, so Talkie today and any later historical model can be scored on the same questions. Nothing is on anyone's critical path, each piece is usable on its own, and Anaël keeps a minimal version on Talkie so the demonstration holds even if a piece arrives late.

The common ground

All projects share one cutoff discipline, train or test only on material earlier than the chosen date. All write their documents into Kaile's canonical record, so provenance is traceable and the corpus is the same one the Time Atlas reads. This is what turns seven sheets into one result.

Rebuilding a Vintage LLM from scratch

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, MARC YOUNES

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build the laboratory's first vintage model on a bounded source and period, then wrap it in a harness that makes it useful, so the past is not only modelled but can be questioned.

Motivation

A vintage model is a generative language model, meaning a model that produces text word after word, trained only on documents earlier than a chosen date, the cutoff. After that cutoff it has seen nothing. Building it is the first half of your work. The second half is the harness, the tooling around the model that turns a weak generator into something usable, retrieval of period documents so it answers from evidence, rewriting of prompts into the language of the time so it understands them, generation by branches and by log-probabilities rather than a single collapsed answer, and, where needed, the support of a stronger model. This is the question of how to make a vintage model actually useful, which no other project in the group covers.

The laboratory already owns Talkie, but a fresh model with a mastered cutoff and source gives the group a new base, the one the others probe, evaluate and stress. And a model alone reasons weakly, so the harness is what lets it be interrogated at all. Building the model and making it usable are the two things the rest of the group builds on.

Sources

Chronicling America, more than twenty one million digitised pages of American newspapers, held by the Library of Congress, entirely in the public domain, so free for research and for a product. Starting slice, the press earlier than 1920. A French window through Gallica or Impresso stays possible late in the semester.

Access, the API and bulk data, chroniclingamerica.loc.gov/about/api, the OCR text, chroniclingamerica.loc.gov/ocr, and Gallica, gallica.bnf.fr.

What to build on

For the model, StoriesLM-v2 gives an encoder baseline that cites the laboratory, suproteem.is/writing/storieslm, the paper of Grave et al. gives temporally ordered pretraining with code, arxiv.org/abs/2605.22769, and the Ranke project gives a historical model family, github.com/DGoettlich/history-llms. For the harness, the moves are retrieval, period-language prompting, branches and log-probabilities, and Talkie's own tooling is a precedent.

Compute

The model has to be trained rather than adapted, for one reason. A genuine vintage model cannot be obtained by fine-tuning a modern model on old text, because the modern base has already seen

everything after the cutoff and that later knowledge leaks back in. Temporal isolation is the whole point, so the model has to be trained from scratch on bounded text. This is affordable because the scale is small, a few hundred million parameters on a bounded slice fits the laboratory GPUs over a semester. Confirm the compute allocation on RCP early.

The steps

1. Frame the cutoff and the source scope, a dated slice of Chronicling America earlier than 1920.
2. Clean and order the corpus in the group's common format, each document with its date, source and rights.
3. Train a first small decoder model from scratch on that slice, small by design.
4. Wrap a first harness around it, retrieval of period documents and prompt rewriting into period language, and show one task the harness makes the model solve that it fails without.

How your work connects with the others

Nobody is blocked if your model arrives late, since your teammates can work on Talkie meanwhile. Once your model exists it becomes their target, and your harness is what makes it answerable, so both halves reinforce the group without the group depending on them.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Selected bibliography

- Levine, Nick, David Duvenaud, and Alec Radford. 2026. "Introducing Talkie: a 13B vintage language model from 1930." https://talkie-lm.com/introducing-talkie. This is the reference vintage model your harness wraps, a starting point rather than a model to retrain.
- He, Songrun, et al. 2025. "Chronologically Consistent Large Language Models." SSRN Scholarly Paper. doi:10.2139/ssrn.5159615. https://papers.ssrn.com/abstract=5159615. It sets the training standard your model must meet, a whole genealogy that respects the cutoff rather than a late filter.

- Kelly, Bryan T., et al. 2026. "Scaling Point-in-Time Language Models." NBER Working Paper. doi: 10.3386/w35247. https://www.nber.org/papers/w35247. It gives the scaling behaviour of point-in-time models, so you size yours honestly for one semester.
- Göttlich, Daniel, et al. 2025. "History LLMs, Ranke-4B." https://github.com/DGoettlich/history-llms. An existing historical model family you can build the harness on instead of training from zero.
- Yan, Yutong, et al. 2026. "DatedGPT: Preventing Lookahead Bias in Large Language Models with Time-Aware Pretraining." arXiv:2603.11838. doi:10.48550/arXiv.2603.11838. https://arxiv.org/abs/2603.11838. A concrete pretraining recipe that keeps the future out, useful if you train rather than reuse.
- Gao, Xin, et al. 2025. "Can Prompts Rewind Time for LLMs? Evaluating the Effectiveness of Prompted Knowledge Cutoffs." EMNLP 2025, pp. 20777 to 20788. doi:10.18653/v1/2025.emnlp-main.1049. https://aclanthology.org/2025.emnlp-main.1049/. It shows that prompting a modern model to act period-bound fails, the argument for a real model plus a real harness.

Key resources

- Chronicling America API
- Chronicling America OCR bulk text
- StoriesLM-v2, historical encoder baseline
- Grave et al., temporally ordered pretraining
- Ranke, history-llms
- Previous DHLAB student projects, all past semesters
- Zoom link

Verifying the cutoff and measuring what the model can do

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, JOE EL HAYEK

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Prove that a vintage model truly ignores the future by detecting leaks at every stage of the pipeline, and, as a second axis, measure what the model can actually do, so the group knows both that its models are clean and what they are capable of.

Motivation

A vintage model is trained only on text earlier than the cutoff, but the declared cutoff is a rule for building the corpus, not a proof of ignorance, because later information slips in through wrong dates, later editions, modern prefaces, merged collections and extraction artefacts. This slipping in is leakage, and measuring it is the core of your project, discussion point 7. The second axis, lighter, is capability, discussion point 1, what the model can do at all, how it learns a task from a few examples, follows an instruction, or uses a tool, and above all why it fails. Your project is the group's validation, both that the models are clean and that their abilities are honestly measured.

Even Talkie, built with an anachronism filter, kept residual knowledge of the New Deal, and TypewriterLM still detected post-cutoff recall after filtering, so a cutoff must be audited, not assumed. And early tests show vintage models reason weakly, leaning on textual patterns, with revealing failure cases such as confusing a name given in the prompt. The group must know what its models truly ignore and truly can do before it builds forecasting or evaluation on them.

Sources

Not a text corpus, but two batteries you build. For leakage, probes on people, events, institutions, technologies and terminology that appeared after the cutoff. For capability, fixed prompt batteries for in-context learning, instruction following and tool use. Both apply to the group's model, to Talkie and to a modern witness. The entities and tasks come from free public sources, so there is no rights constraint.

What to build on

The validity frame is a table of checks, plus the diagnostic ladder, direct recall, then a pre-cutoff pseudo-forecast, then a genuine post-cutoff forecast. Two filtering strategies are worth comparing, the aggressive filtering of Machina Mirabilis, michaelhla.com/blog/machina-mirabilis, and the stratified sampling of Ranke, github.com/DGoettlich/history-llms. For capability, a failure-case approach applies, analyse the causes rather than a single aggregate score.

The steps

1. Build a dated behavioural probe battery, one entity per item with its date of appearance.

2. Measure post-cutoff recall across models, Marc's, Talkie, a modern witness, and map where leaks enter.
3. Set the diagnostic ladder and show that direct recall, pseudo-forecast and genuine forecast are not interchangeable.
4. Build the capability batteries, measure them and analyse the failure cases.
5. Deliver a leak report and a cutoff certificate, plus a capability report with its failure analysis.

How your work connects with the others

You can start immediately on Talkie, without waiting for Marc's model. This project is the group's single validation owner, both integrity and ability, and your certification and numbers are inputs the others may cite or ignore, so you reinforce every project without blocking any.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Selected bibliography

- Sarkar, Suproteem K., and Keyon Vafa. 2024. "Lookahead Bias in Pretrained Language Models." SSRN Scholarly Paper. doi:10.2139/ssrn.4754678. https://papers.ssrn.com/abstract=4754678. The foundational statement of the problem your audit targets, knowledge of the future bleeding in.
- Cheng, Jeffrey, et al. 2024. "Dated Data: Tracing Knowledge Cutoffs in Large Language Models." arXiv:2403.12958. doi:10.48550/arXiv.2403.12958. https://arxiv.org/abs/2403.12958. The method for finding a model's real cutoff, the core of your leakage audit.
- Zhang, Zeyu, Ryan Chen, and Bradly C. Stadie. 2026. "All Leaks Count, Some Count More: Interpretable Temporal Contamination Detection and Mitigation in LLM Backtesting." arXiv: 2602.17234. doi:10.48550/arXiv.2602.17234. https://arxiv.org/abs/2602.17234. A detection and mitigation method you can apply and extend.
- Li, Zehan, et al. 2026. "Simulated Ignorance Fails: A Systematic Study of LLM Behaviors on Forecasting Problems Before Model Knowledge Cutoff." arXiv:2601.13717. doi:10.48550/arXiv. 2601.13717. https://arxiv.org/abs/2601.13717. It shows that asking a modern model to forget the future does not work, which is why a true cutoff matters.

- de Wynter, Adrian. 2025. "Is In-Context Learning Learning?" https://openreview.net/forum?id=bIS0BbYjjd. It frames the central capability question you probe, whether the model reasons or matches patterns.
- Dominguez-Olmedo, Ricardo, Florian E. Dorner, and Moritz Hardt. 2025. "Training on the Test Task Confounds Evaluation and Emergence." ICLR 2025, pp. 68564 to 68594. It warns that apparent capability can be an evaluation artifact, so your probes must guard against it.
- Li, Jiaoda, et al. 2024. "What Do Language Models Learn in Context? The Structured Task Hypothesis." arXiv:2406.04216. doi:10.48550/arXiv.2406.04216. https://arxiv.org/abs/2406.04216. It gives a mechanism for in-context learning you can test on a period model.

Key resources

- Machina Mirabilis, aggressive filtering
- Ranke, history-llms, stratified sampling
- StoriesLM-v2, historical model family
- Previous DHLAB student projects, all past semesters
- Zoom link

The futures the past imagined

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, ELIO KHOURY

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Measure which developments a model of the past regards as plausible from a given date, and compare them with the futures the contemporaries actually imagined, so prediction becomes a genuine historical question rather than a closed quiz.

Motivation

The narrow version of forecasting builds a set of questions open in 1925 whose outcome resolves after, each with a date, a horizon and a resolution rule. That is a solid start, but there is something richer, the futures of the past. Rather than only scoring the model on outcomes we already know, place it at a past date and ask what it treats as likely to come, then set that against what people at the time actually forecast, in their programmes, their press, their patents. The interest is not whether the model guessed right, it is the gap between what was textually probable, what was historically thinkable, and what actually happened.

This is the past-futures idea. It turns a benchmark into a study of how a bounded information state shapes expectation, which is exactly what a model of the past makes newly measurable. The laboratory aims at a first result for Computational Future on 15 and 16 October, and a comparison between the model's imagined futures and the contemporaries' is a striking and legible result for it.

Sources

Impresso, French, German and Luxembourgish newspapers over close to two centuries, with publication metadata and text-reuse clusters, which lets you find both the open debates of the 1920s and the forecasts the contemporaries themselves made. Impresso is used under a research licence. A future product keeps to the free slices.

Access, the project and web app, impresso-project.ch, the Datalab notebooks, github.com/impresso/impresso-datalab-notebooks.

What to build on

A taxonomy of prediction tasks and the rules for writing questions guide you, fix a date, a horizon, a resolution rule, include non-events, respect period language. Existing past-casting sets such as Bench to the Future give models to follow, and the benchmark of Grave et al. gives a scale, arxiv.org/abs/2605.22769. A proper scoring rule, the Brier score, measures a probability against an outcome for the closed part.

The steps

1. In Impresso, find open questions of the 1920s and, alongside, the forecasts the contemporaries made about them.

2. For the closed part, fix a date, a horizon, a resolution rule and a ground truth, and include non-events.
3. Elicit the model's plausible futures from the same date, and compare them with the contemporaries' forecasts and with what happened.
4. Deliver the question set and the comparison, closed scores plus the gap between textually probable, historically thinkable and realised.

How your work connects with the others

You build your set without depending on Marc's model, you can evaluate on Talkie or a published model. Joe's leak certification, when it arrives, strengthens your comparison without being a prerequisite. Your result is a shared evaluation axis for the group and for Computational Future.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Selected bibliography

- FutureSearch, et al. 2025. "Bench to the Future: A Pastcasting Benchmark for Forecasting Agents." arXiv:2506.21558. doi:10.48550/arXiv.2506.21558. https://arxiv.org/abs/2506.21558. It defines pastcasting, the exact setup where the past predicts and the outcome is already known.
- Düring, Marten, et al. 2023. "impresso Text Reuse at Scale." Frontiers in Big Data 6. doi:10.3389/fdata.2023.1249469. https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2023.1249469/full. The Impresso interface that lets you find the debates of 1920 to 1930 that resolve later, the heart of your subject.
- Zou, Andy, et al. 2022. "Forecasting Future World Events With Neural Networks." Advances in Neural Information Processing Systems 35, pp. 27293 to 27305. https://papers.neurips.cc/paper_files/paper/2022/hash/aec870a6772336c15dac992c16f2e7c9-Abstract-Datasets_and_Benchmarks.html. Foundational event forecasting with neural models, the baseline family for your measures.
- Karger, Ezra, et al. 2025. "ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities." arXiv:2409.19839. doi:10.48550/arXiv.2409.19839. https://arxiv.org/abs/2409.19839. A benchmark design that guides how you score a period model's predictions.

- Halawi, Danny, et al. 2024. "Approaching Human-Level Forecasting with Language Models." Advances in Neural Information Processing Systems 37, pp. 50426 to 50468. doi: 10.52202/079017-1598. It shows how far LLM forecasting can go, a reference point for what a vintage model should and should not reach.
- Ye, Chenchen, et al. 2024. "MIRAI: Evaluating LLM Agents for Event Forecasting." arXiv: 2407.01231. doi:10.48550/arXiv.2407.01231. https://arxiv.org/abs/2407.01231. An agentic forecasting evaluation you can adapt to the past.

Key resources

- Impresso project and web app
- Impresso Datalab notebooks
- Grave et al., temporal benchmark
- Previous DHLAB student projects, all past semesters
- Zoom link

The common sense the past never wrote down

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, JITONG ZHANG

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build a benchmark of period common sense and test whether a model of the past holds and can reconstruct the tacit knowledge that its era judged too obvious to write down.

Motivation

Every period has knowledge so ordinary that no one bothers to state it, how a horse is harnessed, what a shilling buys, which illnesses a season brings. Because it was obvious, it is thin in the written record, so a model trained on that record may or may not carry it. This is the most original of the open questions in the field. A benchmark of past common sense, built on the model of SimpleBench but aimed at a period, asks the model questions whose answers a contemporary would have found trivial, and measures whether the model gets them, and whether, when it lacks them, it can reconstruct them from what it does know. Your work is to build that benchmark and run it.

It reframes what a good model of the past is. A model that scores well here holds not just the facts of its period but its unspoken frame, which is what would let it reason as the period did rather than merely quote it. This ties to a deeper idea, that a model which knows little but knows enough to search and reason is a more interesting object than one that has memorised everything, so your benchmark measures exactly the quality that matters.

Sources

Your material is a benchmark you construct, questions of period common sense drawn from period sources, manuals, almanacs, everyday guides, and dialogues, with answers a contemporary would have taken for granted. Draw the period material from free sources, Chronicling America and Gallica for the press and guides, DraCor and the dialogue corpora for everyday exchange. The benchmark itself is your creation, so its rights are clean, cite each source of an item.

Access, Chronicling America, chroniclingamerica.loc.gov/about/api, and Gallica, gallica.bnf.fr.

What to build on

SimpleBench gives the shape of a common-sense benchmark to adapt. The design includes generating the missing common sense synthetically and testing whether training on it helps, and the self-play from period documents attempted on Talkie as a way to build reasoning without leaking the future. You evaluate on Talkie or Marc's model, so no training is needed to start. A well constructed benchmark is a quantitative, self-contained deliverable, which suits your background.

The steps

1. Choose two or three domains of period common sense where the tacit knowledge is rich, for example everyday economic life, health and the seasons, or work and tools.
2. Build the benchmark, questions a contemporary would find trivial, with sourced answers, and non-obvious distractors.
3. Run it on a vintage model and a modern witness, and measure not only the score but whether the model reconstructs a missing answer from adjacent knowledge.
4. Deliver the benchmark and its results, with a first reading of what the model's era makes easy or hard to take for granted.

How your work connects with the others

You return to the vintage group from Temporal to carry this subject, and you work on an already available model, so you depend on no one. Your benchmark is a new evaluation axis for the group, complementary to Elio's futures and Joe's capabilities, and it feeds the shared question of what a good model of the past is. Your earlier work on cross-cutoff surprise stays in reserve for the group, the Temporal project having no student this term.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Selected bibliography

- Chen, Yuqi, et al. 2024. "Surveying the Dead Minds: Historical-Psychological Text Analysis with Contextualized Construct Representation for Classical Chinese." EMNLP 2024, pp. 2597 to 2615. doi:10.18653/v1/2024.emnlp-main.151. https://aclanthology.org/2024.emnlp-main.151/. A method for recovering the mindset of a period from its texts, close to reconstructing tacit knowledge.
- Gao, Lirong, et al. 2026. "Can LLMs Act as Historians? Evaluating Historical Research Capabilities of LLMs via the Chinese Imperial Examination." ACL 2026, pp. 29880 to 29911. doi:10.18653/v1/2026.acl-long.1378. https://aclanthology.org/2026.acl-long.1378/. It uses a period examination as a benchmark, a direct template for your past common sense test.

- Zucchet, Nicolas, et al. 2025. "How do language models learn facts? Dynamics, curricula and hallucinations." arXiv:2503.21676. doi:10.48550/arXiv.2503.21676. https://arxiv.org/abs/2503.21676. It tells you whether implicit knowledge is even acquired in pretraining, which decides if the model can hold tacit knowledge.
- Chang, Hoyeon, et al. 2024. "How Do Large Language Models Acquire Factual Knowledge During Pretraining?" arXiv:2406.11813. doi:10.48550/arXiv.2406.11813. https://arxiv.org/abs/2406.11813. It complements the above on how knowledge enters the model, useful for the synthetic-generation route.
- Varnum, Michael E. W., et al. 2024. "Large Language Models based on historical text could offer informative tools for behavioral science." PNAS 121(42), e2407639121. doi:10.1073/pnas.2407639121. https://www.pnas.org/doi/10.1073/pnas.2407639121. It argues that a period model can serve as an instrument for period norms, the value case for your benchmark.
- Argyle, Lisa P., et al. 2023. "Out of One, Many: Using Language Models to Simulate Human Samples." Political Analysis 31(3), pp. 337 to 351. doi:10.1017/pan.2023.2. https://doi.org/10.1017/pan.2023.2. A method for eliciting period-typical responses, one way to probe unwritten common sense.

Key resources

- Chronicling America API
- Gallica, period guides and press
- Previous DHLAB student projects, all past semesters
- Zoom link

Conversations of the past

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, ISKANDAR GHOBRIL

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. Your exact profile will be confirmed at the kick-off, and the second half of the subject is sized to your level. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Assemble a dated corpus of historical dialogue, then study whose voice the model actually speaks when it talks as the past, and whether less represented voices can be brought out.

Motivation

A vintage model trained on newspapers and books learns to write like the past but not to converse like it, because spontaneous speech was almost never recorded before sound recording arrived around 1900. Old dialogue survives only in written registers close to speech, plays, court testimony, language-learning dialogues. The first half of your work gathers these into one dated corpus, structured by turn and speaker, in the common format. The second half is the question of who we are talking to. A model's corpus is skewed toward a literate, often male perspective, so a user who thinks they speak with someone from 1930 is not speaking with just anyone from 1930. You measure that skew, and you test whether voices that are under represented in the corpus can still be elicited, the n equals all ambition of digital humanities.

An attested dialogue corpus is the material a vintage model needs to gain a conversational register. This register is manufactured by post-training on period speech. Beyond that, the question of whose voice the model carries is one of the more socially interesting in the field, and it is barely studied, so it is a real contribution rather than a reproduction.

Sources

Three constituted sources make the corpus feasible. DraCor, drama in TEI, multilingual, open, dracor.org with its API at dracor.org/doc/api. The Old Bailey Corpus, 1720 to 1913, courtroom speech dated and attributed by speaker, under a research licence, fedora.clarin-d.uni-saarland.de/oldbailey. The Corpus of English Dialogues, 1560 to 1760, under a research licence, varieng.helsinki.fi/CoRD/corpora/CED. The speaker attribution in Old Bailey is what lets you study voice by social position, so it is central to the second half.

What to build on

Much of DraCor and the Old Bailey come with speaker turns, so the corpus work is harmonisation, not transcription. Mr Chatterbox, a Victorian conversational model under MIT licence, is a baseline to probe, huggingface.co/tventurella/mr_chatterbox_model. For the voice study, the silicon-samples literature gives the frame, condition the model on a role by prompting or by period documents, and measure how reliably a less represented voice appears.

The steps

1. Gather and harmonise the three sources into the common format, structured by turn and speaker.
2. Characterise the corpus, volume, periods, languages, registers, and who is represented.
3. Measure the model's default voice, whose perspective it speaks from, and test whether under represented voices can be elicited, with a simple plausibility check that a produced voice is credible for its period.
4. Deliver the dated dialogue corpus and the voice study, kept modest on the evaluation side if your profile calls for it.

How your work connects with the others

You build a corpus, so you depend on no model and no other student. Your corpus enriches the models of the past with a dialogue register, and your voice study gives the group a social reading of what a bounded corpus makes speakable, both additive, neither blocking anyone. The caveat holds, these corpora are too small to pretrain a conversation from scratch, so the use is to fine-tune and then study a model already trained on the large dated slices.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Selected bibliography

- Argyle, Lisa P., et al. 2023. "Out of One, Many: Using Language Models to Simulate Human Samples." Political Analysis 31(3), pp. 337 to 351. doi:10.1017/pan.2023.2. https://doi.org/10.1017/pan.2023.2. The core method for simulating voices from a corpus, and its limits.
- Lin, Zhicheng. 2025. "Six Fallacies in Substituting Large Language Models for Human Participants." Advances in Methods and Practices in Psychological Science 8(3). doi:10.1177/25152459251357566. https://doi.org/10.1177/25152459251357566. A caution list for treating a model as a person, essential before you claim to hear a period voice.
- Beelen, Kaspar, et al. 2025. "Whose news? Critical methods for assessing bias in large historical datasets." Computational Humanities Research 1, e8. doi:10.1017/chr.2025.10007. https://

doi.org/10.1017/chr.2025.10007. It names whose voice the corpus over-represents, the literate and the male, your central problem.

- Shu, Matthew, et al. 2026. "How latent and prompting biases in AI-generated historical narratives influence opinions." PNAS Nexus 5(3), pgag022. doi:10.1093/pnasnexus/pgag022. https://doi.org/10.1093/pnasnexus/pgag022. It shows how a model's historical narrative can shift a reader, the ethical stake of the voices you surface.
- Chen, Yuqi, et al. 2024. "Surveying the Dead Minds: Historical-Psychological Text Analysis with Contextualized Construct Representation for Classical Chinese." EMNLP 2024, pp. 2597 to 2615. doi:10.18653/v1/2024.emnlp-main.151. https://aclanthology.org/2024.emnlp-main.151/. A method for giving analytic voice to historical minds from their texts.
- Levine, Nick, David Duvenaud, and Alec Radford. 2026. "Introducing Talkie: a 13B vintage language model from 1930." https://talkie-lm.com/introducing-talkie. The object itself, the model a user believes speaks from 1930, which your subject interrogates.

Key resources

- DraCor drama corpus
- DraCor API
- Old Bailey Corpus 2.0
- Corpus of English Dialogues
- Mr Chatterbox baseline
- Previous DHLAB student projects, all past semesters
- Zoom link

The first rung of the rediscovery ladder

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, DIEGO PARRA RUIZ

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build the instrument that makes a claim of machine rediscovery defensible, and use it to measure the first rung of the ladder, whether a model that has seen the world only up to 1930 can reach a later discovery once it is handed the analogy that historically enabled it.

Motivation

A vintage model is a generative language model trained only on documents earlier than a chosen date, its cutoff. Everything after that date is, for the model, the future. This makes an old question testable for the first time. Given only what was known before, can a system reach an idea that entered the record later.

One experiment has come close and it defines your starting point. Larraz and Corma reconstruct the information environment before 1936 and ask a modern model to solve the problem that fluid catalytic cracking solved in 1942. With minimal guidance it reaches an FCC-class solution in ten per cent of trials. **Given one cross-domain analogy, the pneumatic handling of grain, it succeeds in every trial.** The evidence was already available, and the bottleneck was the connection between two domains.

That experiment has one flaw. Their model was pretrained long after 1942, so it may simply have recognised the answer. A vintage base removes that objection. The laboratory has one, Talkie, cut at 1930, and a second is being trained inside this group.

The missing piece is a measuring instrument. There is no protocol that says what counts as a rediscovery case, no scored ladder of hints, no rule that separates having the idea from retrieving it. Without that, any success is an anecdote and any failure proves nothing. Building it is your project.

Read this next sentence carefully, because it defines what you will be graded on. Your deliverable is the instrument and the curve, not a discovery. A benchmark on which no model rediscovers anything is a valid and publishable result, provided you can say at which rung of hinting the behaviour changes. You are not being asked to gamble a semester on whether a model succeeds.

What the ladder is

The working report of this group sets out four rungs, and you work on the first, with the second as an extension if time allows.

1. **Given the key idea.** Can the model use a supplied analogy or precursor to reach the solution.
2. **Find the key idea.** Can it identify the useful analogy among several candidates.
3. **Build the solution.** Can it combine historical evidence to reconstruct a later discovery.
4. **Discover autonomously.** Can it identify an open problem and produce a plausible solution.

Your hint ladder is the graded version of rung one, no hint, domain hint, analogy named, analogy explained, with the success rate measured at each level. **The scientific object is the shape of that curve**, the jump from ten to one hundred in the FCC precedent, not the endpoint.

Sources

One constraint shapes everything here. **A case is admissible only if the discovery postdates the model's cutoff.** With Talkie at 1930 this excludes continental drift, Snow and cholera, antisepsis, the benzene ring, the whole nineteenth century. Your window is roughly the 1930s to the 1950s, which is exactly where the FCC case sits.

Three criteria for retaining a case. An analogical transfer documented by historians of science rather than reconstructed after the fact. All ingredients demonstrably available before the cutoff, on dated sources. And a solution a non-specialist can score, failing which the evaluation becomes a thesis in the history of science.

For the pre-cutoff evidence, the group's corpus and the dated collections it reads, Chronicling America for the press, the patent record for techniques and devices, and the periodical literature of the field concerned. The report suggests four families of target, patents, mathematical techniques, experimental methods and scientific theories, chosen at different distances after the cutoff.

Selecting the cases is the first month and it is the decision that determines the semester. It is made with Frédéric, not alone.

Hindsight

The protocol comes before the runs, for one reason. You will build the cases knowing the answers. This is the structural flaw of every rediscovery benchmark and it is not cured by being careful, it is cured by a procedure.

Each case you write is given blind to another student in the group, who attempts it without knowing what is being sought. If a human solves it too easily, the hint has leaked into the statement of the problem. This costs half an hour of the Tuesday slot and it earns a full section of your report.

And every run needs a modern control. A current model succeeding without a hint is measuring leakage, not discovery. Prompting a modern model to behave as though it were period-bound is not a control either, and the literature shows it fails.

The steps

1. Write the case protocol, target discovery and date, pre-cutoff evidence set, enabling analogy, decoy analogies, resolution rule.
2. Build the FCC case first, as calibration, since its outcome is published, and reproduce the published result with a vintage base rather than a modern one.
3. Build three to four fresh cases in the 1930 to 1950 window, on dated sources, and blind-test each one inside the group.
4. Run the hint ladder on Talkie, on the group's model when it arrives, and on a modern control, with a contamination check per case.
5. Report the curve, the failure modes, and the cases the instrument could not score.

How your work connects with the others

You consume rather than produce infrastructure. Marc's harness is how you query the model, and Talkie is your fallback until it exists. Joe's leakage certificate is what makes your positive results defensible, and you are his most demanding client, since a rediscovery claim without a contamination check is worth nothing. Elio's dated question set and yours share a format, so keep them compatible. Anaël may take one of your cases as a vignette for the October demonstration, but nothing of his depends on you, and nothing of yours depends on him.

Nobody is blocked if your instrument arrives late, and if it works the whole group inherits it.

Key dates

- Kick-off, Tuesday 8 September 2026, already held. Slides and group briefing will be sent to you.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- **First working hour on the list of cases, to be fixed with Frédéric for 22 September**, since that choice decides the semester.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, supervisor Frédéric Kaplan. Confirm beforehand the number of credits your study plan expects and whether Tec de Monterrey validates a semester project here.
- Weekly progress email, every Friday, a short message with what you did, what you found and what blocks you. Standing rule for the whole group.

Selected bibliography

- Larraz, R., and A. Corma. 2026. "Human analogical guidance amplifies LLM performance through cross-domain knowledge activation." *Nature Communications* 17 (1): 4822. doi:10.1038/s41467-026-70873-7. The experiment you extend, and the source of the ten to one hundred jump. Read it for its design and for the flaw you correct, its model postdates the discovery it is asked to find.
- Hla, Michael. 2026. "Machina Mirabilis." https://michaelhla.com/blog/machina-mirabilis.html. GPT-1900 specialised for physics reasoning, with partial successes near discrete energy quanta and the equivalence of gravity and acceleration, and with regressions worth studying. The honest record of how hard rung three is.
- Donini, Anaël. 2026. *Vintage LLMs, working report.* Internal, EPFL/VintageLM/. Section on scientific rediscovery, where the four-rung ladder is set out and the benchmark families are proposed. Your subject is the operationalisation of that section.
- Gentner, Dedre. 1983. "Structure-Mapping: A Theoretical Framework for Analogy." *Cognitive Science* 7 (2): 155 to 170. The theory that lets you build a decoy set on principle rather than by intuition, by distinguishing surface similarity from shared relational structure. Your decoys should be surface-close and structurally wrong.

- Langley, Pat, Herbert Simon, Gary Bradshaw, and Jan Zytkow. 1987. *Scientific Discovery, Computational Explorations of the Creative Processes.* MIT Press. The ancestor of your project, programs that rediscovered Kepler and Ohm from data, and the critique that sank them, the task had been pre-digested by the people who built it. That critique is the one you must design against.
- Swanson, Don R. 1986. "Undiscovered Public Knowledge." *The Library Quarterly* 56 (2): 103 to 118. The thesis, forty years before FCC was rerun, that discoveries can lie latent in a literature that already contains all their parts, unconnected. The intellectual charter of the whole exercise.
- He, Songrun, et al. 2025. "Chronologically Consistent Large Language Models." SSRN. doi: 10.2139/ssrn.5159615. The cutoff standard your models must meet, a genealogy that respects the date rather than a filter applied late.
- Gao, Xin, et al. 2025. "Can Prompts Rewind Time for LLMs? Evaluating the Effectiveness of Prompted Knowledge Cutoffs." EMNLP 2025, 20777 to 20788. https://aclanthology.org/2025.emnlp-main.1049/. Why asking a modern model to pretend it stopped in 1930 is not a control, which is what forces the vintage base and the modern control to be two separate arms.

Key resources

- Talkie, the 1930 model
- Machina Mirabilis, GPT-1900
- Chronicling America API
- Espacenet, historical patent record
- Previous DHLAB student projects, all past semesters
- Zoom link

Building the Pastcasting Benchmark

SEMESTER PROJECT, VINTAGE LANGUAGE MODELS, ANAËL DONINI

Vintage Language Models group, DHLAB, autumn 2026. Supervision Frédéric Kaplan. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build the Pastcasting Benchmark. It places a model of the past before a cutoff and measures what that model treats as likely to come against what actually happened. The demonstration shown at Computational Future is its first public instantiation, assembling the pieces the other Vintage students produce into one working object.

Motivation

A model of the past is only interesting if we can measure what it knows and what it can anticipate. The past is the only place where the answers are already known. Pastcasting is exactly that test: the model is placed at a given date and asked what it expects, then scored against what actually happened. **Your project turns pastcasting into a reusable scientific benchmark.** The demonstration at Computational Future on 15 and 16 October is its first public instantiation. Framing it this way is what makes the work survive October, since what remains after the event is the instrument rather than the screen. Because you continue the summer work on Talkie, you start from a real base while the other projects are still taking shape, which is what lets the demonstration stand on time.

What you integrate

Your benchmark is built from the others' work, not in parallel to it. Marc's model and its harness give you the system under test and the tooling to query it. Joe's leakage certificate is what makes the result credible, without a guarantee that the model did not see the future, a good forecast proves nothing. Elio's set of questions open at a past date and resolved later is the core material of the benchmark, and the contrast he studies, what the past imagined against what came, is the narrative of the demonstration. Jitong's past common sense and Iskandar's recovered voices give two short vignettes that show the breadth of the model beyond forecasting. You assemble these into one measured, legible result.

Beyond integrating the group's outputs, you define the evaluation protocol itself, how questions are represented, how forecasts are scored, how uncertainty is shown, and how the benchmark can later be extended to other periods and other models.

The benchmark is independent of any particular language model. Talkie is the system under test today, the group's own model joins it this term, and TypewriterLM or a future historical model can be run through the same questions and the same scoring without rebuilding anything. That independence is what makes the object reusable and comparable, and it is the reason to design the format carefully now.

Sources

Talkie, the existing 1930 vintage model, is your system under test, and the group's own model as it matures is the second. Impresso is the source for questions debated around 1920 and 1930 whose outcome resolves later, and Chronicling America gives an English complement. The pastcasting and forecasting benchmark literature gives you the scoring frame to follow rather than invent.

Access, the Impresso Datalab, github.com/impresso/impresso-datalab-notebooks, Chronicling America, chroniclingamerica.loc.gov/about/api, and the Swiss AI initiative behind the infrastructure, swiss-ai.org.

What to build on

You do not invent the evaluation from nothing. Existing pastcasting benchmarks, notably Bench to the Future, and the dynamic forecasting benchmarks such as ForecastBench, give you the question format and the scoring rules, a date, a horizon, a resolution rule, a proper score such as the Brier score. Talkie and the summer work give you a model already deployable, and the other Vintage projects give you the harness, the leakage audit and the question set, so your effort is integration and measurement rather than rebuilding each part.

The steps

1. With Elio, fix a first set of questions open around 1925 and resolved after, each with a date, a horizon and a resolution rule.
2. Wrap the model through Marc's harness and apply Joe's leakage certificate, so the runs are both usable and credible.
3. Run the pastcasting, score the model against the known outcomes, and record the gap between the textually probable, the historically thinkable and what actually happened. Write the protocol so a second model can be scored on the same questions without change.
4. Add the two vignettes from Jitong and Iskandar, and package the whole as the Computational Future demonstration, with a minimal version that stands on Talkie alone as the safety net.

How your work connects with the others

You are the point where the Vintage group becomes one result. Each student hands you a piece, the model and harness, the leakage certificate, the question set, the two vignettes, and you compose them into the benchmark and the demonstration. You sit on the others' outputs, so you coordinate rather than block, and your minimal version on Talkie means the demonstration holds even if a piece arrives late.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.

- Impresso sessions of the FDH course, week 4, Wednesday 30 September 13:00 to 15:00 on the platform and its search, and Thursday 1 October 10:00 to 12:00 on its processing chain, directly relevant to your question set.
- Computational Future, Thursday 15 and Friday 16 October 2026, where your benchmark is the demonstration.
- Weekly supervision, your slot is Tuesday 10:00 to 11:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Vintage Language Models, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Impresso Datalab notebooks
- Chronicling America newspaper API
- Swiss AI initiative
- Previous DHLAB student projects, all past semesters
- Zoom link

PROJECT GROUP

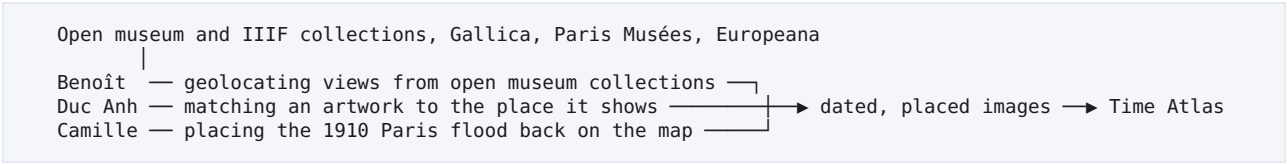
Geolocating Artworks

3 sheets, preceded by the articulation page

How the Geolocating Artworks projects articulate

DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Isabella di Lenardo. One schematic page on how the Geolocating projects fit together. Weekly slot, Tuesday 9:15 to 10:00, room INN 128.

The picture



The links

The three projects share one task, to put a visual document back in the place it represents, and they differ by angle. Benoît geolocates views drawn from open museum collections at scale. Duc Anh works the harder matching problem, from an artwork to the exact place it depicts. Camille takes one event, the 1910 Paris flood, and rebuilds its map from an image corpus, so her work is the event-centred case of the same method.

They reinforce each other through shared sources and shared matching techniques, and none blocks the others, each can progress on its own collection. Isabella co-supervises the group and holds the Tuesday block, which is why the group presents first.

The common ground

Every result is a dated, placed record that lands in the Time Atlas through the same record model the rest of the cohort uses, so a geolocated artwork sits on the same map as a newspaper article or a heritage page. The image sources are reached through the same IIIF door the IIIF group builds.

Geolocating artworks from open museum collections

SEMESTER PROJECT, GEOLOCATING ARTWORKS, BENOÎT MEYLAN

Geolocating Artworks group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Isabella di Lenardo. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Assemble a corpus of artworks whose depicted place is real and nameable, drawn from open museum collections, and build a baseline that predicts the location a view shows.

Motivation

Geolocating an artwork means attaching to it the real place it depicts, as a coordinate, not a caption. The clean case is the urban view, a painting, print or drawing of a named street, square, monument or skyline, because its location is a real place with a ground truth to check against. Your work is to gather many such views from collections that are open, meaning free to reuse, attach a verified place to each, and then build a first model that, given a view, proposes where it is. That prediction baseline is the heart of the group's problem, put on a corpus you control.

A located artwork is a point the Time Atlas can show in space and time, so your corpus feeds the atlas directly, and your baseline is the reference the rest of the group measures against. Working from open collections means the result can travel beyond the semester, into teaching and into a product, without a rights problem.

Sources

Your core is open museum collections that publish under CC0 or the public domain and expose their images through IIIF, the interoperability standard. The Art Institute of Chicago offers a clean public API with CC0 images, and the Rijksmuseum publishes its collection in the public domain with a full IIIF service. The Met, the Getty and the Harvard Art Museums add more of the same. Widen the pool with two large freely licensed sets, the British Library book illustrations, more than a million public-domain images mostly from nineteenth-century books, and the laboratory's own Wikimedia Commons collection assembled in a previous semester project, available internally. For ground truth, GeoNames and Wikidata turn a place name into a coordinate.

Access, Art Institute open access and API, artic.edu/open-access/public-api, Rijksmuseum IIIF, data.rijksmuseum.nl/docs/iiif, the IIIF museums group for the others, iiif.io/community/groups/museums, British Library book images, huggingface.co/datasets/biglam/british-library-book-images, and Wikimedia Commons, commons.wikimedia.org.

What to build on

The museum APIs and IIIF services give you the images and their metadata without a custom crawler per institution. Pretrained vision models give you a strong image representation off the shelf, so your baseline can start as retrieval, find the known view most similar to a query, before any training. Image-geolocation research gives you the evaluation frame, distance between

predicted and true location. The genre to target is the urban view, because its ground truth is immediate.

The steps

1. Harvest urban views from the open museum collections, filtered on rights, and widen with the British Library and Wikimedia Commons sets.
2. Attach a verified place to each, from the caption through a gazetteer, checked against Wikidata.
3. Build a first location baseline, retrieval then a simple predictor, and measure it by distance error.
4. Deliver the located corpus in the group's common schema and the baseline with its evaluation.

How your work connects with the others

You work from your own sources, so you depend on no one. Within Geolocating Artworks you sit beside Duc Anh Do, who matches a single artwork to the exact modern place it shows, and Camille Dupré-Tabti, who geolocates the 1910 flood, so the three located corpora share the common format and sit in one map without any dependency. Your baseline is the reference the others can compare their matches against.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab. Especially useful for you, since your corpus is meant for the atlas.
- Weekly supervision, your slot is Tuesday 9:15 to 10:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Geolocating Artworks, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Art Institute of Chicago, open access and API
- Rijksmuseum IIIF
- IIIF Museums community group, Met, Getty, Harvard
- British Library book images, public domain
- Wikimedia Commons
- GeoNames gazetteer
- Previous DHLAB student projects, all past semesters

- Zoom link

Matching an artwork to the place it shows

SEMESTER PROJECT, GEOLOCATING ARTWORKS, DUC ANH DO

Geolocating Artworks group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Isabella di Lenardo. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Given a historical view or topographic print of a place, find the modern photograph of the same place by visual matching, and produce verified links from artwork to location.

Motivation

This is geolocation by correspondence rather than by caption. A historical print of a street or a monument and a present-day photograph of the same spot share a structure, a building outline, a skyline, a bridge, even across centuries and across the gap between an engraving and a photograph. Matching means building a pipeline that, given the print, retrieves the modern image that shows the same place, and confirms it. The output is a set of artwork-to-place links, each anchored by a real match rather than a guessed coordinate. This is the hardest and most rewarding case of the group's problem, cross-domain visual matching.

A confirmed match turns a historical image into a precisely located point, and a located point is what the Time Atlas needs. Because your match ties the old view to a modern reference with a known coordinate, your links carry a ground truth stronger than a caption alone, which makes your output the sharpest input the atlas can receive from the group.

Sources

Your historical inputs are Gallica topographic prints, the estampes of the Bibliothèque nationale de France, exposed through IIIF, broadly reusable, filter the rights field. Your modern reference is the Google Landmarks dataset, a large public set of photographs of named places, which gives you the target side of the match. Widen both sides with Wikimedia Commons, very large and freely licensed, and with the British Library book illustrations for extra historical views. Confirm each match against Wikidata or GeoNames so the coordinate is verifiable.

Access, Gallica and its IIIF image API, gallica.bnf.fr and api.bnf.fr/api-iiif-de-recuperation-des-images-de-gallica, the Google Landmarks dataset, github.com/cvdfoundation/google-landmark, Wikimedia Commons, commons.wikimedia.org, and the British Library book images, huggingface.co/datasets/biglam/british-library-book-images.

What to build on

Cross-domain matching is an active field with strong tools off the shelf. Local feature matching and image retrieval give you a first pipeline without training, and pretrained vision models and vision-language models give representations that survive the jump from print to photograph. The Google Landmarks work gives you both a dataset and an evaluation frame for place recognition. Benoît Meylan's located corpus in the group is a ready set of views to test your matching against.

The steps

1. Assemble a set of historical topographic prints from Gallica, filtered on rights.
2. Build a matching pipeline, retrieval then local feature verification, against the modern reference photographs.
3. Evaluate the matches, precision of the link and distance error of the resulting coordinate.
4. Deliver the artwork-to-place links in the group's common schema, with the evaluation.

How your work connects with the others

You work from your own sources, so you depend on no one. Within Geolocating Artworks you sit beside Benoît Meylan, whose located corpus gives you views to match, and Camille Dupré-Tabti, who geolocates the 1910 flood, so the three located corpora share the common format and one map without any dependency. Your confirmed matches are the strongest located points the group hands the Time Atlas.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab. Especially useful for you, since your matches feed the atlas.
- Weekly supervision, your slot is Tuesday 9:15 to 10:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Geolocating Artworks, supervisor Frédéric Kaplan.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Gallica topographic prints, IIIF
- Gallica IIIF image API
- Google Landmarks dataset
- Wikimedia Commons
- British Library book images, public domain
- Wikidata, for verifying a place
- Previous DHLAB student projects, all past semesters
- Zoom link

Placing the 1910 Paris flood back on the map

SEMESTER PROJECT, GEOLOCATING ARTWORKS, CAMILLE DUPRÉ-TABTI

Geolocating Artworks group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Isabella di Lenardo. Digital Humanities minor project, 12 ECTS. This sheet describes your subject, your resources, your steps and the key dates of the semester. Your subject is a sub-subject within Geolocating Artworks, the geolocation of a single historical event corpus.

Goal

Harvest the visual record of the January 1910 flood of the Seine, georeference each image to the exact place in Paris it shows, and deliver a mapped, dated and rights-checked corpus of the event.

Motivation

Geolocating an image means attaching to it the precise place it depicts, a street, a bridge, a square, as a coordinate rather than a caption. The 1910 flood, la grande crue de la Seine, was one of the most photographed events of its time, so the surviving record is dense, dated to the day, and tied to identifiable Paris locations. Your work is to turn that scattered record into a corpus where every image carries its date, its place as a coordinate, its source and its rights, so the flood can be read across the map of the city rather than one picture at a time. This is the same geolocation problem the group works on, applied to one event and one city.

The event gives the group a dense, well-dated and highly localisable corpus, which is the ideal case for testing geolocation on historical images. It also carries an environmental and heritage reading, how a city recorded a disaster, that connects directly to the spatio-temporal atlas, where a located and dated image becomes a point the map can show in time. Your sub-subject is distinct from your neighbours in the group, who geolocate artworks and views from museum and print collections, while you geolocate the photographic record of a single event.

Sources

Your primary source is Paris Musées Collections, in particular the Musée Carnavalet, which holds a large open set of photographs and postcards of the 1910 flood, many under open content, meaning free to reuse. Widen it with Europeana, which aggregates flood material from many institutions, where rights vary item by item and must be filtered before use. Check the rights field on every item, keep the open content for anything that might leave the lab, and cite each source.

Access, Paris Musées Collections, parismuseescollections.paris.fr, and Europeana, europeana.eu.

What to build on

Geolocation of historical images is the shared method of your group, so start from the baselines your neighbours Benoît and Duc Anh are setting up rather than from nothing. For placing an image on the map, a gazetteer, meaning a dictionary of place names with coordinates, turns a caption such as rue de Charenton into a point, and GeoNames covers modern Paris well. For serving and comparing heritage images at scale, the IIIF standard is the group's common door, and georeferenced map collections such as David Rumsey show how a historical image is anchored to

coordinates. Many Carnavalet captions already name the street or the arrondissement, which gives you a strong starting signal before any automatic matching.

The steps

1. Harvest the 1910 flood imagery from Paris Musées Collections, filtered on rights, and widen with Europeana where the licence allows.
2. Georeference each image to its place, first from the caption through a gazetteer, then by visual matching where the caption is vague.
3. Build the corpus in the group's common format, each image carrying its date, its coordinate, its source and its rights.
4. Map the result, and where possible bring it into the Time Atlas so the flood can be read across the city in time.

How your work connects with the others

You work from your own source and event, so you depend on no one. Within Geolocating Artworks you sit beside Benoît Meylan and Duc Anh Do, and you share the geolocation approach and the common format with them, which lets the three located corpora sit in one map without any dependency between you. Your corpus also feeds the Time Atlas directly, adding a dated and located event to it.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab. Especially useful for you, since your corpus is meant for the atlas.
- Weekly supervision, your slot is Tuesday 9:15 to 10:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, project Geolocating Artworks, supervisor Frédéric Kaplan, co-supervisor Isabella di Lenardo, confirmed at the kick-off.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Paris Musées Collections, Carnavalet
- Europeana
- GeoNames gazetteer
- IIIF standard

- David Rumsey georeferenced maps
- Previous DHLAB student projects, all past semesters
- Zoom link

PROJECT GROUP

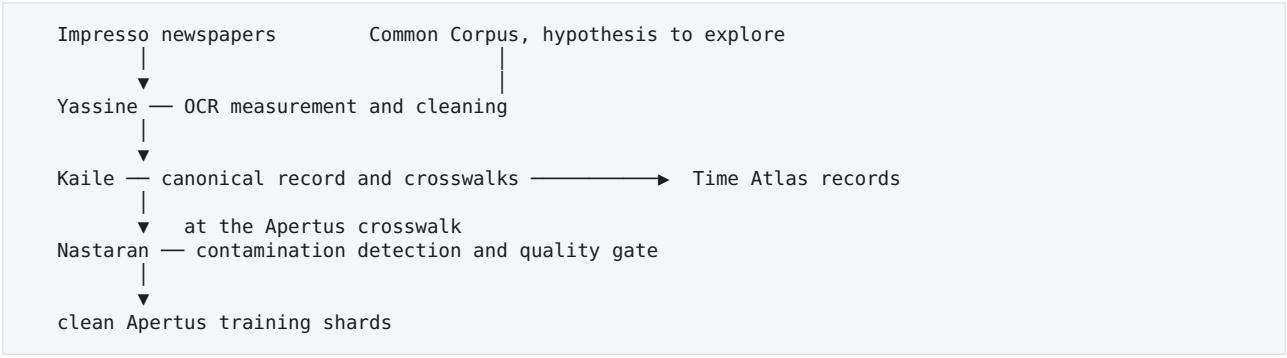
Preparing Historical Data for Apertus

3 sheets, preceded by the articulation page

How the Apertus projects articulate

DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Maud Ehrmann. One schematic page on how the Apertus projects fit together. Weekly slot, Tuesday 11:30 to 12:00, room INN 128.

The picture



The links

Yassine measures and improves the optical character recognition of the press, so the text that enters the pipeline is as clean as it can be. Kaile is the pivot, he defines one canonical record and the crosswalks that move a document between the Impresso model, the Apertus training shards and the Time Atlas, minting one identifier that travels across all three. Nastaran owns the trust, her contamination detection and her data-quality evaluation form the gate that Kaile's Apertus crosswalk calls, so only dated-true, rights-clean material is emitted.

The division is clean and deliberate. Kaile owns the representation and the plumbing, Nastaran owns the judgment of what is allowed through, Yassine feeds clean text in. They advance together while staying separately gradable, the pipeline on one side and the guarantee on the other.

The common ground

The canonical record is the shared coordinate system, and the same prepared document is at once trainable by Apertus and plottable in the Time Atlas. Common Corpus is a source to test, not a settled choice. Whatever is declared dated is checked to be dated, which is the quiet guarantee the whole cluster depends on.

Measuring and cleaning the OCR

SEMESTER PROJECT, PREPARING HISTORICAL DATA FOR APERTUS, YASSINE DAFER

Preparing Historical Data for Apertus group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Maud Ehrmann. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Measure the quality of the text that optical character recognition produces from digitised historical newspapers, and build a cleaning pipeline that raises that quality enough to make the material worth training on.

Motivation

Optical character recognition, OCR, is the step that turns the image of a scanned page into machine-readable text. On historical newspapers it is noisy, broken words, wrong letters, columns read in the wrong order, and that noise is not cosmetic. A model learns far less from noisy transcription than from clean text, so the value of a historical corpus for a model like Apertus depends directly on how clean its OCR is. Your work is to measure how bad it is, on real material, and to make it measurably better.

Apertus is the Swiss open large language model, and the group prepares historical data for it. The single result that frames your project is blunt, conventionally transcribed historical text was found to deliver only about a third of the learning efficiency of clean transcription, rising to roughly two thirds after simple cleaning. That gap is the value you can recover. Every point of OCR quality you win is training signal the model would otherwise lose, so your measurement and your cleaning are what turn a large but noisy archive into usable data.

Sources

Gallica, the digitised collections of the Bibliothèque nationale de France, exposes a very large body of historical press with OCR already produced and, for part of it, already corrected, which gives you both raw and improved text to compare. Widen with Impresso, the French, German and Luxembourgish newspaper collection, which carries rich metadata and text-reuse links. Gallica press is broadly reusable, Impresso is under a research licence, so check the rights field before publishing any derived corpus.

Access, Gallica and its IIIF image API, gallica.bnf.fr and api.bnf.fr/api-iiif-de-recuperation-des-images-de-gallica, and the Impresso Datalab, github.com/impresso/impresso-datalab-notebooks.

What to build on

You are not building an OCR engine, you are measuring and repairing its output. Character and word error rate against a small hand-checked reference give you the measurement, and Gallica's corrected press gives you a ready reference to score raw OCR against. For cleaning, start simple, the regular-expression and normalisation passes that already recover a large share of the loss, before reaching for a model-based correction. Impresso's own OCR work and its text-reuse

detection show how duplicated wire copy inflates a corpus and must be handled separately from cleaning.

The steps

1. Sample a defined slice of Gallica press, raw OCR and, where available, its corrected version.
2. Measure quality, character and word error rate against a small hand-checked reference, and report it by period and by title.
3. Build a cleaning pipeline, normalisation and correction, and measure the quality it recovers.
4. Deliver a cleaned, quantified corpus in the group's common schema, with the before-and-after numbers that show the gain.

How your work connects with the others

You work from your own source, so you depend on no one. On Apertus you sit beside Kaile Chu, who owns the common schema, so your cleaned text and her schema meet in the same prepared dataset, your cleanliness filling the fields she defines. A clean, dated corpus is also what the language models of the past need, so your gain reaches the vintage and temporal projects without any dependency between you.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Impresso sessions of the FDH course, week 4, Wednesday 30 September 13:00 to 15:00 on the platform and its search, and Thursday 1 October 10:00 to 12:00 on its processing chain, directly relevant to your source.
- Weekly supervision, your slot is Tuesday 11:30 to 12:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, Preparing Historical Data for Apertus, supervisor Frédéric Kaplan, co-supervisor Maud Ehrmann.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Gallica, Bibliothèque nationale de France
- Gallica IIIF image API
- Impresso Datalab notebooks

- Swiss AI, the Apertus initiative
- Previous DHLAB student projects, all past semesters
- Zoom link

The contamination and data-quality gate for the prepared corpus

SEMESTER PROJECT, PREPARING HISTORICAL DATA FOR APERTUS, NASTARAN HASHEMISANJANI

Preparing Historical Data for Apertus group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Maud Ehrmann. This is your Apertus option, one of the two subjects offered to you while you decide. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build the contamination detection and the data-quality evaluation, the judgment layer that decides whether a prepared historical corpus is clean and dated-true enough to train Apertus, and plug it in as the gate of the pipeline that Kaile builds, so that only trustworthy material reaches the model.

Motivation

Apertus is the Swiss open large language model, and the group prepares historical data for it. Two defects make a dated corpus untrustworthy. Contamination is material that does not belong, a document whose date is wrong, a text that carries later material, a piece of wire copy duplicated across many titles and inflating the record. Poor data quality is text too broken to learn from, bad OCR, boilerplate, fragments. Your subject is to catch both, a contamination detection that flags what is mis-dated or anachronistic, and a data-quality evaluation that scores whether a prepared slice is clean enough to use. This is where your LLM-as-a-Judge background applies directly, a model that reads a candidate document and rates its fitness.

The articulation with Kaile is the clean division that makes Apertus work. He owns the representation and the plumbing, the canonical record and the crosswalks that move a document between Impresso, the Apertus training shards and the Time Atlas. You own the trust, the checks that decide what is allowed through. Your gate plugs into his Apertus crosswalk, so together you deliver one prepared dataset, the pipeline on his side and the guarantee on yours, with neither owning the other's half. Without your layer the Apertus cluster would ship a corpus that looks dated but is not, which is the quiet failure your evaluation exists to prevent.

Sources

Common Corpus, assembled by Pleias, is a hypothesis to explore rather than a settled choice, a large multilingual training corpus carrying date, language, collection and source metadata, with an open-culture subset in the public domain. Impresso adds French, German and Luxembourgish newspapers with rich publication metadata and text-reuse clusters, under a research licence, and its text-reuse clusters are a direct handle for catching duplicated wire copy. You run your checks on slices of these as they flow through the pipeline.

Access, Common Corpus on Hugging Face, huggingface.co/datasets/PleIAs/common_corpus, Impresso Datalab, github.com/impresso/impresso-datalab-notebooks, and the Swiss AI initiative behind Apertus, swiss-ai.org.

What to build on

You do not define the schema, you read the canonical record Kaile specifies and judge what fills it, so your effort goes into the checks, not the format. For contamination, text-reuse detection and deduplication are established techniques, and temporal-validity checks, does the vocabulary, the named entities and the references fit the claimed date, give you the anachronism signal. For quality, your own LLM-as-a-Judge work is the base, a judge prompted to score a document on legibility, coherence and period fit, calibrated against a small hand-labelled set. Yassine Dafer, beside you on Apertus, measures and improves OCR quality, so his signal is one input your evaluation consumes rather than recomputes.

The steps

1. Define the contamination checks, wrong date, later material, and wire-copy duplication, each producing a flag with a reason, on the canonical record Kaile circulates.
2. Build the data-quality evaluation, an LLM-as-a-Judge that scores a document on legibility, coherence and period fit, calibrated on a small hand-labelled sample.
3. Run both on an Impresso and, if it is used, a Common Corpus slice flowing through the pipeline, and report how much is flagged and why.
4. Deliver the detection and quality method, a quality report on the slice, and the gate that Kaile's Apertus crosswalk calls, with a cleaned slice as the result.

How your work connects with the others

Your articulation with Kaile is the spine of Apertus, he pipes and you judge, his crosswalk calls your gate, and the two of you hand over one trustworthy prepared dataset. You depend on his canonical record to read, and he depends on your gate to emit clean shards, so you advance together while staying separately gradable. Yassine's cleaned OCR feeds the same dataset, his cleanliness and your judgment meeting on the same documents. The language models of the past train only on what your gate lets through, so your guarantee reaches the whole Apertus cluster.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Impresso sessions of the FDH course, week 4, Wednesday 30 September 13:00 to 15:00 on the platform and its search, and Thursday 1 October 10:00 to 12:00 on its processing chain, directly relevant to your source.
- Weekly supervision, your slot is Tuesday 11:30 to 12:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.

- IS-Academia registration to complete, Preparing Historical Data for Apertus, supervisor Frédéric Kaplan, co-supervisor Maud Ehrmann. Credits, 12 per your M4 study plan, confirmed by Frédéric on 8 September, scope set to match.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Common Corpus on Hugging Face
- Impresso Datalab notebooks
- Swiss AI, the Apertus initiative
- Previous DHLAB student projects, all past semesters
- Zoom link

The pivot between Apertus ingestion, Impresso, and the Time Atlas

MASTER PROJECT, PREPARING HISTORICAL DATA FOR APERTUS, KAILE CHU

Preparing Historical Data for Apertus, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Maud Ehrmann. Master project, PDM, so the scope is larger and the deliverable more complete than a semester project. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Build the canonical representation and the two-way crosswalks that let a single historical document live at once as training data ingested for Apertus, as an Impresso record, and as a Time Atlas record, carrying one stable identifier, a verified date, a place, and propagated rights, so that three data worlds that do not naturally speak to each other become one pipeline.

Motivation

Three systems describe the same historical documents in three different shapes, and today a document crosses from one to another only by hand. The way sources are ingested for Apertus is a flat, large-scale view, a document is text plus the few fields training needs, a date, a source, a language and a licence, deduplicated and filtered for contamination. Impresso is the opposite, a richly structured newspaper model, issues and pages and articles, publication metadata, named entities, OCR quality and text-reuse clusters, held under a research licence. The Time Atlas is a third shape again, a spatio-temporal record model where every object is a typed record placed in space and time, each with a stable identifier, a date, a place, and cross-links to other records. Each shape is right for its purpose and none reads the others.

Your master project is the pivot between the three. You define one canonical record that sits in the middle, and the mappings that turn an Impresso article into that canonical record, the canonical record into a clean Apertus training shard, and the same canonical record into a Time Atlas entry, where a dated and placed document becomes a record cross-linked to the places and events it touches. The prize is that a newspaper article becomes, in one pass, both material Apertus can train on and a dated placed point the atlas can plot, keeping the same identifier throughout so the training shard, the Impresso article and the atlas record always refer back to one another. This is the keystone that makes the past computable across the language models and the map, and it is why the role is a full master project rather than a schema on a page.

The three systems you connect

The canonical record is the heart of the work. It holds the fields every system needs, an identifier, a date with its precision, a place, a language, the rights, the source, and the text-reuse cluster where known, and it is typed so it can specialise, a newspaper article, a book page, a corpus. Around it you build three crosswalks. The Impresso crosswalk reads the newspaper model and fills the canonical record, carrying the publication date, the place of publication, the entities and the reuse cluster, and recording where the OCR is weak. The Apertus crosswalk writes the canonical record out as a training shard, flat text with its date, language and licence, deduplicated, and gated so that contaminated or mis-dated material does not pass, the gate itself being the contamination

and quality layer that Nastaran builds, which you plug in rather than reinvent. The Time Atlas crosswalk writes the canonical record as a typed record placed in space and time with its cross-links, so the document lands on the map without a second conversion. The identifier is the thread, it is minted once and travels across all three, which is what lets a result in the atlas point back to the exact training document and the exact newspaper article.

Sources

Common Corpus, assembled by Pleias, is a hypothesis to explore rather than a settled choice, a very large multilingual training corpus carrying date, language, collection and source metadata, with an open-culture subset in the public domain, so where it is free it is usable for training, and it may serve as a model for what an Apertus-ready shard looks like, to be confirmed. Impresso adds French, German and Luxembourgish newspapers with rich publication metadata and text-reuse clusters, under a research licence, and it is the richest of your three inputs, so it is the natural source to drive the pivot. The Time Atlas contributes the spatio-temporal record model that your canonical record aligns to. Chronicling America gives a free English newspaper source to test the pivot on a second collection.

Access, Common Corpus on Hugging Face, huggingface.co/datasets/PleIAs/common_corpus, Impresso Datalab, github.com/impresso/impresso-datalab-notebooks, the Swiss AI initiative behind Apertus, swiss-ai.org, and Chronicling America, chroniclingamerica.loc.gov/about/api.

What to build on

You do not design in a vacuum, you reconcile models that already exist. Common Corpus, should it be used, offers a field set for training data, Impresso already exposes a documented data model with publication metadata and text-reuse clusters, the Time Atlas already structures its records in space and time, and the IIIF manifest already standardises date, place and rights for heritage objects. Your canonical record is the least common multiple of these, and your crosswalks are field-to-field mappings you specify and then implement. Crosswalk and ontology-alignment practice gives you the method. Yassine Dafer, beside you on Apertus, measures and improves OCR quality, so his output is a direct input to your Impresso crosswalk, and Nastaran Hashemisanjani builds the contamination and data-quality layer that your Apertus crosswalk calls as its gate.

The steps

1. Map the three data models side by side and produce a crosswalk table, field by field, naming what matches, what is missing and what must be derived.
2. Specify the canonical record and a stable identifier scheme, with a controlled vocabulary for date precision, place and rights, and circulate it so the group writes into it from week one.
3. Build the working pivot on a slice, Impresso into the canonical record, then out to an Apertus training shard on one side and to Time Atlas records on the other, keeping the shared identifier across all three.
4. Plug in Nastaran's contamination and quality gate at the Apertus crosswalk, so only dated-true, rights-clean material is emitted, and propagate the rights field end to end.
5. Deliver the adopted specification, the two-way pipeline on the slice, and an evaluation of its coverage, with a cleaned Apertus-ready dataset that is already a set of Time Atlas records.

How your work connects with the others

You are the keystone of the Apertus cluster and its bridge to the atlas, and your articulation with Nastaran is the clean division that makes Apertus work, you own the representation and the plumbing, the canonical record, the crosswalks and the identifiers, while she owns the contamination detection and the data-quality judgment that decide whether what flows is trustworthy. Her gate plugs into your Apertus crosswalk, so the two of you deliver one prepared dataset, pipeline on your side and guarantee on hers, without either owning the other's half. The language projects become additive because they read and write your canonical record, and they can start on a provisional version this week while you harden it, so you reinforce every one of them without sitting on anyone's critical path. Yassine's cleaned OCR flows into your Impresso crosswalk, and the geolocation and image projects land their records in the same atlas through the same record model.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Impresso sessions of the FDH course, week 4, Wednesday 30 September 13:00 to 15:00 on the platform and its search, and Thursday 1 October 10:00 to 12:00 on its processing chain, directly relevant to your pivot.
- Weekly supervision, your slot is Tuesday 11:30 to 12:00 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete as a master project, PDM, on Preparing Historical Data for Apertus, supervisor Frédéric Kaplan, co-supervisor Maud Ehrmann.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- Common Corpus on Hugging Face
- Impresso Datalab notebooks
- Swiss AI, the Apertus initiative
- Chronicling America newspaper API
- IIIF standard, for date, place and rights fields
- Previous DHLAB student projects, all past semesters
- Zoom link

PROJECT GROUP

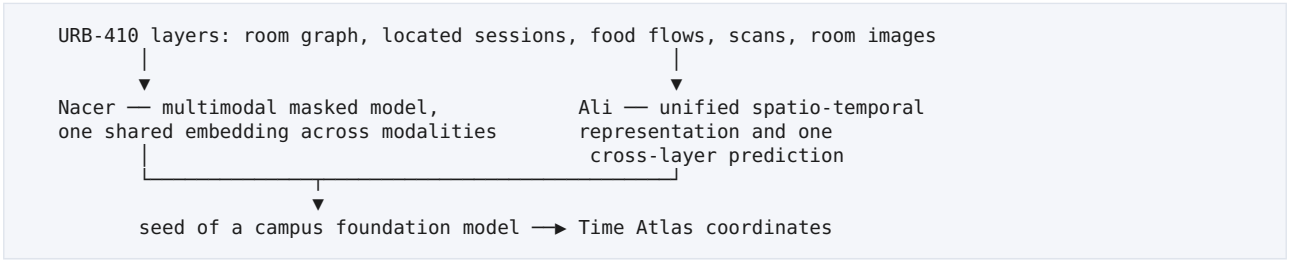
EPFL Digital Twin Foundation Model

2 sheets, preceded by the articulation page

How the EPFL Digital Twin projects articulate

DHLAB, autumn 2026. Supervision Frédéric Kaplan, assistant Alexander Rusnak. One schematic page on how the Digital Twin projects fit together. Weekly slot, Tuesday 12:00 to 12:30, room INN 128.

The picture



The links

The two projects are the two halves of the world-models seed. Ali joins the existing campus layers into one representation keyed on space and time and proves it with a single cross-layer prediction, for example occupancy from schedule or food demand from occupancy. Nacer learns the shared multimodal embedding, a masked model that fuses geometry, occupancy, flows and room images into one space where a place at a moment is a single point.

They compose, Nacer's embedding can feed Ali's prediction, while neither sits on the other's critical path, Ali holds a baseline to start alone and Nacer can begin on two modalities. Both reuse the URB-410 prototype rather than collecting anew, with Rusnak in support.

The common ground

A campus placed in space and time speaks the same coordinates as the Time Atlas and the common record, so the world-models side stays distinct from the language and image projects yet aligns with them on the shared spatio-temporal frame.

A foundation model for the campus

SEMESTER PROJECT, EPFL DIGITAL TWIN FOUNDATION MODEL, ALI TAHRI HASSANI

EPFL Digital Twin group, DHLAB, autumn 2026. Supervision Frédéric Kaplan, with Alexander Rusnak for the world-models side. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Take the layers that already describe the EPFL campus in space and time, build a first unified representation that connects them, and show one prediction across layers, the seed of a campus foundation model.

Motivation

A digital twin is a living model of a real place, fed by its data, that can be queried and can predict. A foundation model is a single representation trained broadly enough that many tasks plug into it rather than each being solved from scratch. For EPFL the material largely exists already, so your work is not to collect it again but to join it. The URB-410 course produced, in one spring semester, room polygons and a navigation graph for every floor of Écublens, roughly two hundred and ninety thousand course sessions placed in space and time, and about four million food transactions across three years, with time as the dimension that connects them. Your project takes these layers and builds the representation that holds them together, then proves it with one prediction that crosses from one layer to another.

The campus is the one place the laboratory can instrument fully and observe over time, so it is the natural testbed for a spatio-temporal foundation model, the world-models side of the group's research. The existing prototype has the spatial substrate, the temporal regime and the interfaces, but the layers still sit side by side rather than as one model. Turning them into a single representation that predicts across layers is the step that moves the effort from a set of demonstrations toward an actual foundation model, and it is a scoped, honest contribution for one semester.

Sources

Your material is the existing campus data, not new collection. The room polygons and the navigation graph, the located course sessions from the IS-Academia occupancy export, and the food transactions from RESCO, all already prepared in the URB-410 prototype. Work from the aggregated and processed layers, not from raw individual traces, because the prototype already handles presence and transaction records that touch individuals, so consent and access are a live question you must respect and record rather than assume. As a spatial complement, the RoomAgents scans give volumetric geometry for the rooms already captured.

Access to these layers is arranged with the laboratory and the URB-410 team at the kick-off.

What to build on

The four URB-410 projects of spring 2026 are your direct foundation. One built the spatial substrate, room polygons, a Dijkstra navigation graph and cohort-level movement between

classes. One built food-flow prediction with LightGBM, reaching an out-of-sample coefficient of determination near 0.79, and made the argument that time is the layer that connects everything. RoomAgents built a spatial representation by Gaussian splatting with a semantic backbone distilling CLIP, SigLIP, SAM and DINOv2, and a conversational agent grounded in the room. A fourth collected room photographs through a game. Your representation reuses these rather than rebuilding them, and the LightGBM flow model gives you a baseline to beat or to extend across layers.

The steps

1. Get access to the existing layers and load them, room polygons and navigation graph, located sessions, food flows, and the RoomAgents scans.
2. Define a unified representation keyed on space and time, so a place at a time carries its occupancy, its flows and its geometry together.
3. Build one cross-layer predictive proof, for example predicting room occupancy from schedule and cohorts, or food demand from occupancy, and evaluate it against the existing baseline.
4. Deliver the unified representation, the cross-layer prediction with its evaluation, and a clear note on data access and consent.

How your work connects with the others

You are the seed of the world-models side, distinct from the language and image projects but sharing the same spatio-temporal frame, so a campus placed in space and time speaks the coordinates of the Time Atlas and of the common schema. You sit closest to Alexander Rusnak's students on 3D and city-scale questions, and your representation extends the URB-410 prototype toward the foundation model the SwissAI grant aims at, without depending on any other semester project.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, Tuesday in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September, your exact slot confirmed at the first session.
- Weekly supervision, your slot is Tuesday 12:00 to 12:30 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, EPFL Digital Twin Foundation Model, supervisor Frédéric Kaplan, assistant Alexander Rusnak, to confirm.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- **Urban Digital Twins, the course textbook by Frédéric Kaplan**, 240 pages, the reference for the architecture in seven components, the scenario engine and the hybrid models. PDF, Direct link. Course page, URB-410, Moodle, go.epfl.ch/URB-410
- **Final presentations of the URB-410 students, 2026**, Direct link. The four prototype projects presented in full, the spatial substrate and student flows, the food flows on RESCO transactions, RoomAgents on Gaussian splatting, and the alternate reality game. Start here rather than from a blank page, and say what you extend
- **Camille Lannoye**, camille.lannoye@epfl.ch, contact for the EPFL Digital Twin model. She built the spatial substrate of the URB-410 prototype, the room polygons of every floor at Écublens, the single navigation graph over corridors, stairs and lifts, and the student flow simulation on four academic years of IS-Academia occupancy. Ask her for access to those layers rather than rebuilding them
- The URB-410 prototype and its datasets, room graph, located sessions, food flows, RoomAgents scans, internal to the laboratory
- The room polygons and navigation graph of the Écublens site
- Previous DHLAB student projects, all past semesters
- Zoom link

A multimodal model of the campus

SEMESTER PROJECT, EPFL DIGITAL TWIN FOUNDATION MODEL, NACER SAADANI-HASSANI

EPFL Digital Twin group, DHLAB, autumn 2026. Supervision Frédéric Kaplan, with Alexander Rusnak for the world-models side. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Train a multimodal masked model of the EPFL campus that learns one shared embedding across its very different signals, so that a place at a moment becomes a single point you can query and translate from one modality to another.

Motivation

The campus is described by signals of different natures, the geometry of the rooms and their scans, the schedules and the occupancy, the food flows over three years, and photographs of the spaces. Today these live side by side. Your work is to learn one representation that holds them together, a masked multimodal model in the spirit of the 4M and nano4M approach you already practised, trained to predict any missing modality from the others. The result is a shared embedding space where geometry, time, occupancy and image sit on the same coordinates, so a query in one modality can retrieve or reconstruct another. This is the representation backbone of the twin, distinct from and complementary to the predictive layer.

A digital twin becomes a foundation model only when its heterogeneous layers share one representation rather than being joined case by case. Learning that shared embedding is the step that turns a set of separate datasets into something a single model can reason over. Your background fits it precisely, since training a model from scratch with masked multimodal objectives is what nano4M gave you, and here you apply it to a real place you have lived in for three years.

Sources

Your material is the existing campus data, not new collection. The room polygons and the navigation graph, the located course sessions from the IS-Academia occupancy export, the food transactions from RESCO, the RoomAgents scans for volumetric geometry, and the room photographs collected through the URB-410 game. Work from the aggregated and processed layers rather than raw individual traces, because occupancy and transaction records touch individuals, so consent and access are a live question to respect and record. Filter each layer on its access terms before training.

Access to these layers is arranged with the laboratory and the URB-410 team at the kick-off.

What to build on

The four URB-410 projects of spring 2026 are your direct foundation, the spatial substrate with room polygons and a navigation graph, the food-flow model, and above all RoomAgents, which already built a spatial representation by Gaussian splatting with a semantic backbone distilling CLIP, SigLIP, SAM and DINOv2. Your masked multimodal model reuses these encoders as its per-modality front ends rather than rebuilding them, and the nano4M and 4M code give you a training

frame that exists. Ali Tahri Hassani, beside you, builds the unified representation and one cross-layer prediction, so agree early on the shared coordinates you both write into.

The steps

1. Get access to the existing layers and load at least two of different natures, for example located sessions and room geometry.
2. Define per-modality encoders, reusing the RoomAgents semantic backbone where it fits, and a shared token or embedding space keyed on space and time.
3. Train a first masked multimodal model on a bounded slice, learning to predict a held-out modality from the others.
4. Evaluate it, cross-modal retrieval and reconstruction, and show one any-to-any query working, then report the failure cases.

How your work connects with the others

You and Ali are the two halves of the world-models seed, your shared embedding and his unified representation and prediction, composing into one twin while staying separately gradable. Neither is on the other's critical path, you can start on two modalities before his representation is complete, and your embedding can later feed his prediction. A campus placed in space and time speaks the coordinates of the Time Atlas and of the common schema, so your representation aligns with the rest of the cohort without depending on it.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, Tuesday in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September, your exact slot confirmed at the first session.
- Weekly supervision, your slot is Tuesday 12:00 to 12:30 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, EPFL Digital Twin Foundation Model, supervisor Frédéric Kaplan, assistant Alexander Rusnak, to confirm.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- **Urban Digital Twins, the course textbook by Frédéric Kaplan**, 240 pages, the reference for the architecture in seven components, the scenario engine and the hybrid models. PDF, Direct link. Course page, URB-410, Moodle, go.epfl.ch/URB-410
- **Final presentations of the URB-410 students, 2026**, Direct link. The four prototype projects presented in full, the spatial substrate and student flows, the food flows on RESCO transactions, RoomAgents on Gaussian splatting, and the alternate reality game. Start here rather than from a blank page, and say what you extend
- **Camille Lannoye**, camille.lannoye@epfl.ch, contact for the EPFL Digital Twin model. She built the spatial substrate of the URB-410 prototype, the room polygons of every floor at Écublens, the single navigation graph over corridors, stairs and lifts, and the student flow simulation on four academic years of IS-Academia occupancy. Ask her for access to those layers rather than rebuilding them
- The URB-410 prototype and its datasets, room graph, located sessions, food flows, RoomAgents scans and room photographs, internal to the laboratory
- The 4M and nano4M masked multimodal modelling approach, your training frame
- Previous DHLAB student projects, all past semesters
- Zoom link

PROJECT GROUP

IIIF Document Information Extraction

2 sheets, preceded by the articulation page

How the IIIF projects articulate

DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Ziyi Liu. One schematic page on how the IIIF projects fit together. Weekly slot, Tuesday 11:00 to 11:20, room INN 128.

The picture



The links

Boheng is the front of the pipeline, he censuses the institutions that publish through IIIF and harvests their manifests by rights, so a scattered patrimony arrives through one door. Hafidz is the reading half, he runs optical character recognition on the harvested page images and repairs it with a language model, then measures the error rate. Ziyi's own research reads the page image directly without a text step, so his OCR-free engine is the sibling approach, and comparing the two readings on the same pages is part of Hafidz's subject.

The two student halves compose into one pipeline while staying separately gradable, Hafidz can start on a public sample before the harvest is complete, and the harvest is useful even before extraction runs.

The common ground

Every extracted record lands in the common canonical record and flows into the Time Atlas, and the harvesting door Boheng builds is the same one the Geolocating projects use to reach museum and library images. The IIIF group is the connective infrastructure of the image side.

One door to a scattered heritage

SEMESTER PROJECT, IIIF DOCUMENT INFORMATION EXTRACTION, BOHENG SHANG

IIIF group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Ziyi Liu, whose PhD builds an OCR-free search engine for IIIF resources. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Census the institutions that publish heritage through IIIF and build the unified harvest that pulls their records across providers, filtered on rights, so a scattered patrimony arrives through one door in the common schema.

Motivation

IIIF, the International Image Interoperability Framework, is the standard libraries and museums use to publish their images. Its unit is the manifest, the record that describes one object and carries its images and its metadata of date, place and rights. Because so many institutions expose IIIF, one access method can reach across all of them, hundreds of millions of objects, without recoding a crawler per provider. Your half of the project is the front, discover the providers and their manifests, harvest them by rights, and deliver dated, placed, rights-checked records into the group's common schema. Your teammate takes the second half, reading the content of what you harvest.

This is the connective infrastructure of the image side. When the harvest exists, the geolocation projects reach museum and library images through one door instead of many, and the Time Atlas receives dated and located heritage as a stream rather than a series of one-off imports. It aligns with Europeana and the Heritage consortium where the laboratory is already engaged.

Sources

The aggregators come first, because they expose many providers at once. Europeana publishes IIIF manifests at the scale of tens of millions of objects and offers a Change Discovery API, the harvesting interface, and the IIIF registry lists providers' activity streams, the natural entry point for a census. Biblissima aggregates manuscripts and rare books. The reservoirs follow, Internet Archive and Gallica for digitised pages, the Library of Congress, David Rumsey for georeferenced maps, and the open museums. Every manifest carries a rights statement, so filter on it before harvesting.

Access, IIIF Change Discovery, iiif.io/api/discovery/1.0, the IIIF registry, registry.iiif.io, Biblissima, iiif.biblissima.fr/collections, Gallica IIIF, api.bnf.fr/api-iiif-de-recuperation-des-images-de-gallica.

What to build on

Ziyi's project is the anchor and the IIIF discovery APIs give harvesting without a custom crawler per provider. The reported image-count survey already gathered in the laboratory gives a realistic sense of scale and of how uneven the metadata is. Your neighbour on the extraction half reads what you harvest, so agree early on the record format you both use, the common schema.

The steps

1. Census the IIIF providers through the registry and Europeana's Change Discovery API, and record their scale and rights.
2. Build a unified harvest, discover and pull manifests filtered on rights, without a per-provider access.
3. Normalise each record into the common schema, its identifier, date, place, rights and source.
4. Deliver the provider census and a harvested, schema-conformant slice ready for extraction and for the Time Atlas.

How your work connects with the others

You harvest, your teammate Hafidz Arsha extracts, so the two halves compose into one pipeline while staying separately gradable. Neither is on the other's critical path, extraction can start on a public sample before your harvest is complete, and your harvest is useful even before extraction runs. Your records land in Kaile's common schema and flow into the Time Atlas, and the image projects reach their sources through your door.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 11:00 to 11:20 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, IIIF Document Information Extraction, supervisor Frédéric Kaplan, co-supervisor Ziyi Liu.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- IIIF standard, the framework itself
- IIIF Change Discovery API
- IIIF registry of activity streams
- Biblissima IIIF collections
- Gallica IIIF image API
- Previous DHLAB student projects, all past semesters
- Zoom link

Reading heritage with OCR

SEMESTER PROJECT, IIIF DOCUMENT INFORMATION EXTRACTION, HAFIDZ ARSHA

IIIF group, DHLAB, autumn 2026. Co-supervision Frédéric Kaplan and Ziyi Liu, whose PhD builds an OCR-free search engine for IIIF resources. This sheet describes your subject, your resources, your steps and the key dates of the semester.

Goal

Extract the content of the heritage page images harvested through IIIF by running and improving optical character recognition, cleaning its output, and building a searchable extracted slice.

Motivation

A harvested IIIF manifest carries an object, its images and its metadata, but the usable text is often thin, a bad OCR layer or none at all. Your half of the project turns those page images into clean text. You run optical character recognition on them, then you correct and evaluate what comes out, because raw OCR on historical typefaces and degraded scans is noisy. Modern engines and language-model post-correction change what is achievable here, so the work is to build a pipeline that reads a page, recognises its text, repairs the recognition, and writes a searchable record. Ziyi's research takes the opposite route, reading the page image directly without a text step, so your OCR pipeline and his OCR-free engine are two siblings on the same material, and comparing them on the same pages is part of your subject. Your teammate takes the first half, harvesting the manifests you read.

Extraction is what turns a pile of harvested images into something queryable. OCR remains the workhorse that produces actual text a search can index and a historian can read, and its quality on old print is the practical bottleneck of every downstream use. Measuring that quality honestly, and pushing it up with post-correction, is a concrete and gradeable contribution, and setting it side by side with the OCR-free approach tells the group where each one wins.

Sources

Your primary input is the harvested slice from your teammate on the discovery half, but you do not wait for it, you can start on a public sample of page images from Gallica or Internet Archive. The tools you use are open, established OCR engines such as Tesseract and Kraken, the handwriting and historical stack of eScriptorium and Transkribus, and transformer OCR such as TrOCR, with a language model for post-correction of the recognised text. Filter each source on its rights before use.

Access, Gallica IIIF, api.bnf.fr/api-iiif-de-recuperation-des-images-de-gallica, Internet Archive, archive.org, the Tesseract engine, github.com/tesseract-ocr/tesseract, and Kraken with eScriptorium, kraken.re.

What to build on

You do not build an OCR engine from scratch, you assemble and tune existing ones, then add a language-model correction pass, which is where the recent gains are. A useful reference for that exact move is Levchenko, Maria A., 2025, Evaluating LLMs for Historical Document OCR, a

Methodological Framework for Digital Humanities, aclanthology.org/2025.lm4dh-1.7, which frames how a language model helps read a historical page. The Impresso and Gallica pipelines already carry large OCR corpora with known quality issues, so you have realistic material and baselines. Ziyi's OCR-free engine gives you the comparison partner, and his task benchmark gives a shared yardstick. Agree early with your teammate on the common schema you both write into.

The steps

1. Take a tranche of page images, from your teammate's harvest or a public sample to start.
2. Run OCR with a modern engine, then a language-model post-correction pass, and produce clean text per page.
3. Measure the quality, character and word error rates against a small ground truth, and record where old typefaces and degraded scans break it.
4. Write the cleaned text into the common schema as a searchable slice, and compare your OCR result with Ziyi's OCR-free reading on the same pages.

How your work connects with the others

Your teammate Boheng Shang harvests, you read with OCR, so the two halves compose into one pipeline while staying separately gradable. Neither is on the other's critical path, you can start on a public sample before the harvest is complete. Your OCR pipeline and Ziyi's OCR-free engine are the two readings of the same material, and your comparison is what tells the group when to run which. Your extracted records land in the common schema and feed the Time Atlas and the image projects with searchable content.

Key dates

- Kick-off, Tuesday 8 September 2026, 11:15 to 12:30, room INN 128, and on Zoom, epfl.zoom.us/j/2693459226.
- Time Atlas introduction, open sessions from the FDH course, Wednesday 16 September 13:00 to 15:00 in CM1100, Thursday 17 September 10:00 to 12:00 in BC03, Thursday 17 September 13:00 to 15:00 in BC04 for the optional lab.
- Weekly supervision, your slot is Tuesday 11:00 to 11:20 in room INN 128, also on Zoom, epfl.zoom.us/j/2693459226. First session Tuesday 22 September.
- Mid-term presentation, Monday 9 November 2026, all day.
- Report submission, Friday 8 January 2027.
- Final presentations, Monday 18 January 2027.
- IS-Academia registration to complete, IIIF Document Information Extraction, supervisor Frédéric Kaplan, co-supervisor Ziyi Liu.
- Weekly progress email, every Friday, send Frédéric Kaplan a short message with the week's advancement, what you did, what you found and what blocks you. This is a standing rule for the whole group.

Key resources

- IIIF standard, the framework itself
- Tesseract OCR engine

- Kraken and eScriptorium, historical OCR
- Evaluating LLMs for Historical Document OCR, Levchenko 2025
- Gallica IIIF image API
- Internet Archive
- Previous DHLAB student projects, all past semesters
- Zoom link